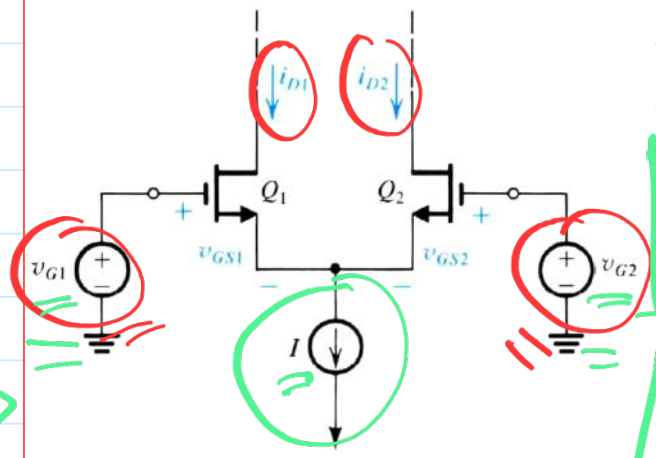
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 9.2 The BJT Differential Pair

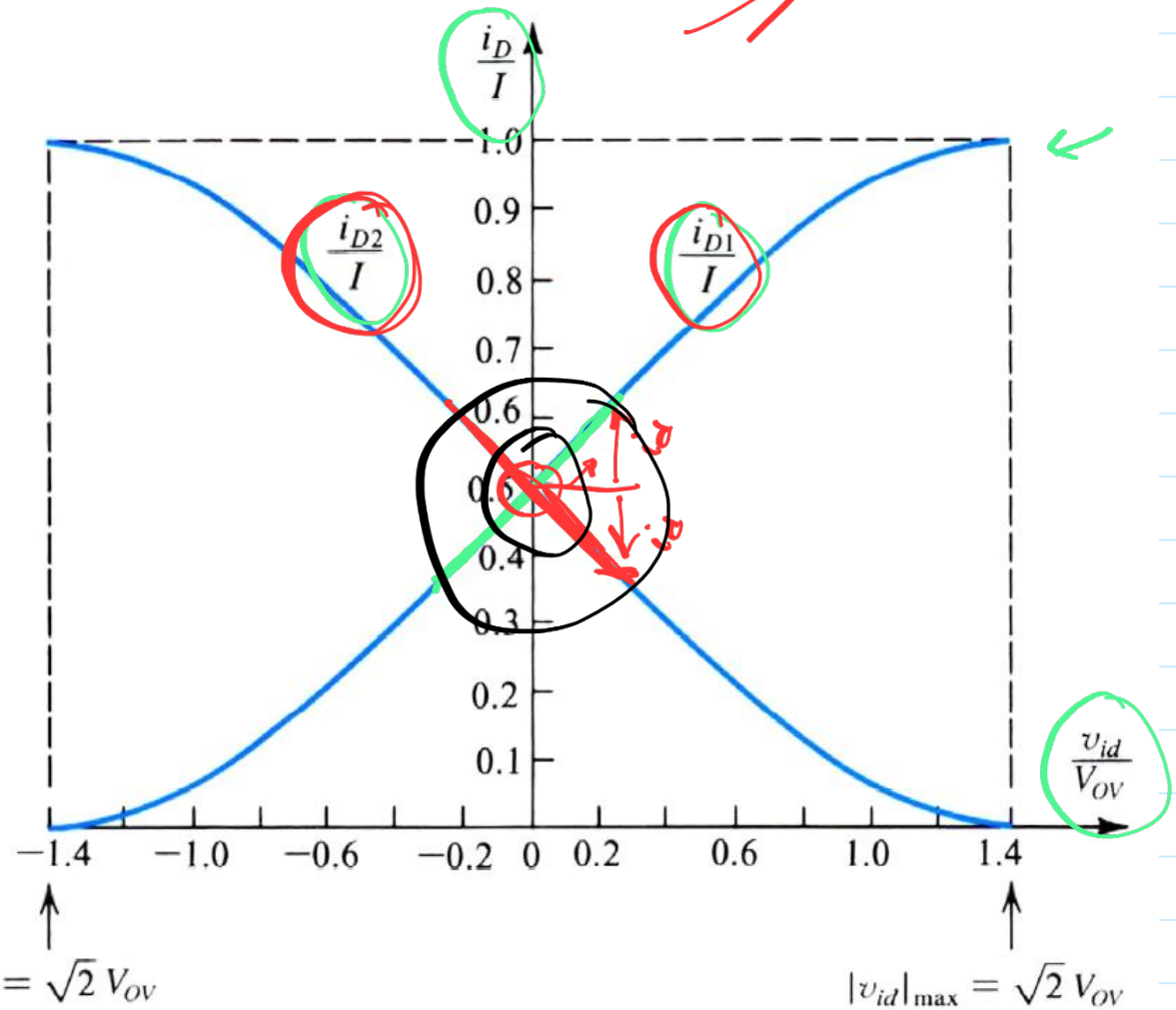
Recap



$$v_{GS1} - v_{GS2} = v_{G1} - v_{G2} = v_{id}$$

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$

$$i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$

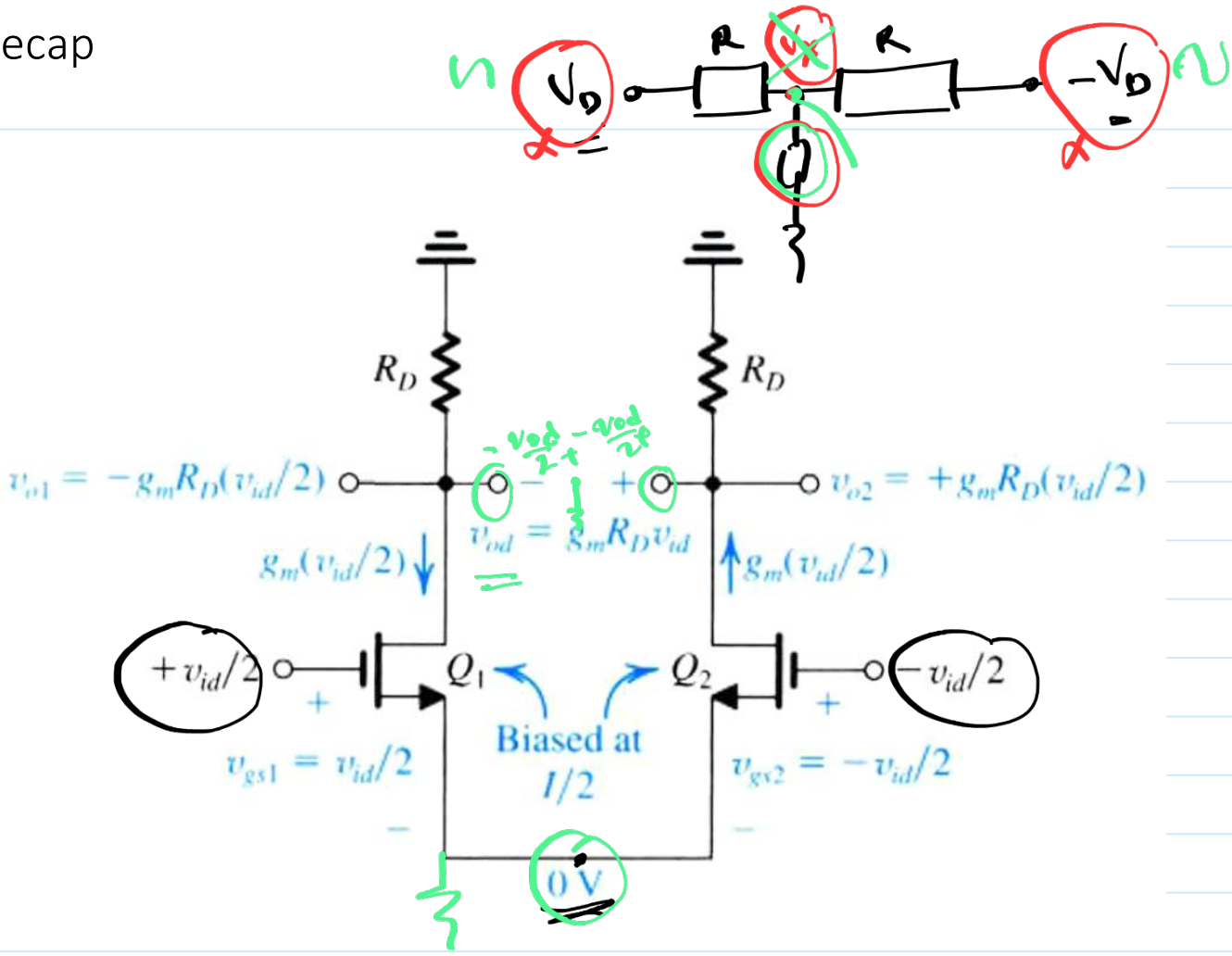


$|v_{id}|_{max} = \sqrt{2}\,V_{OV}$ $\qquad$ $|v_{id}|_{max} = \sqrt{2}\,V_{OV}$

$$i_{D1} \simeq \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)$$

$$i_{D2} \simeq \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)$$

Recap



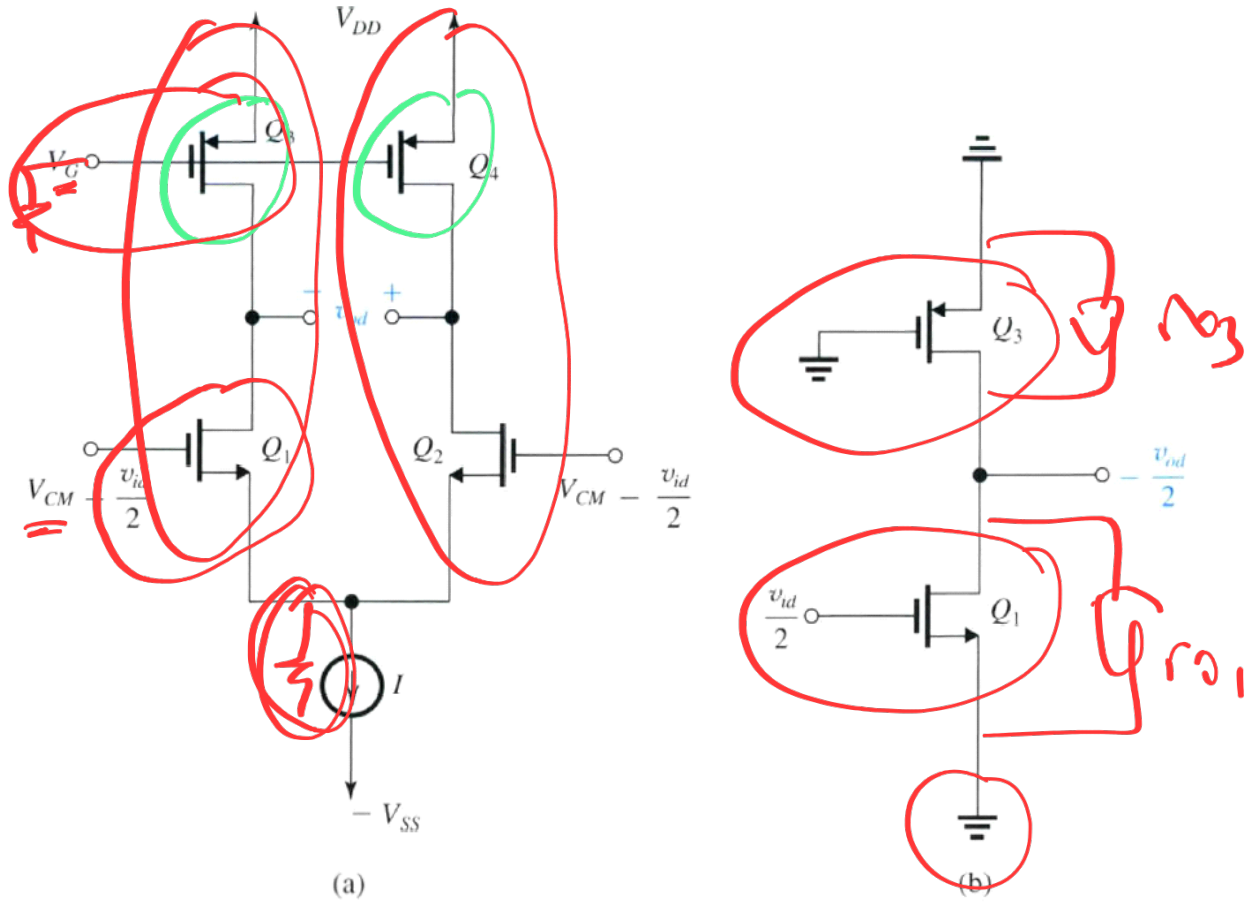
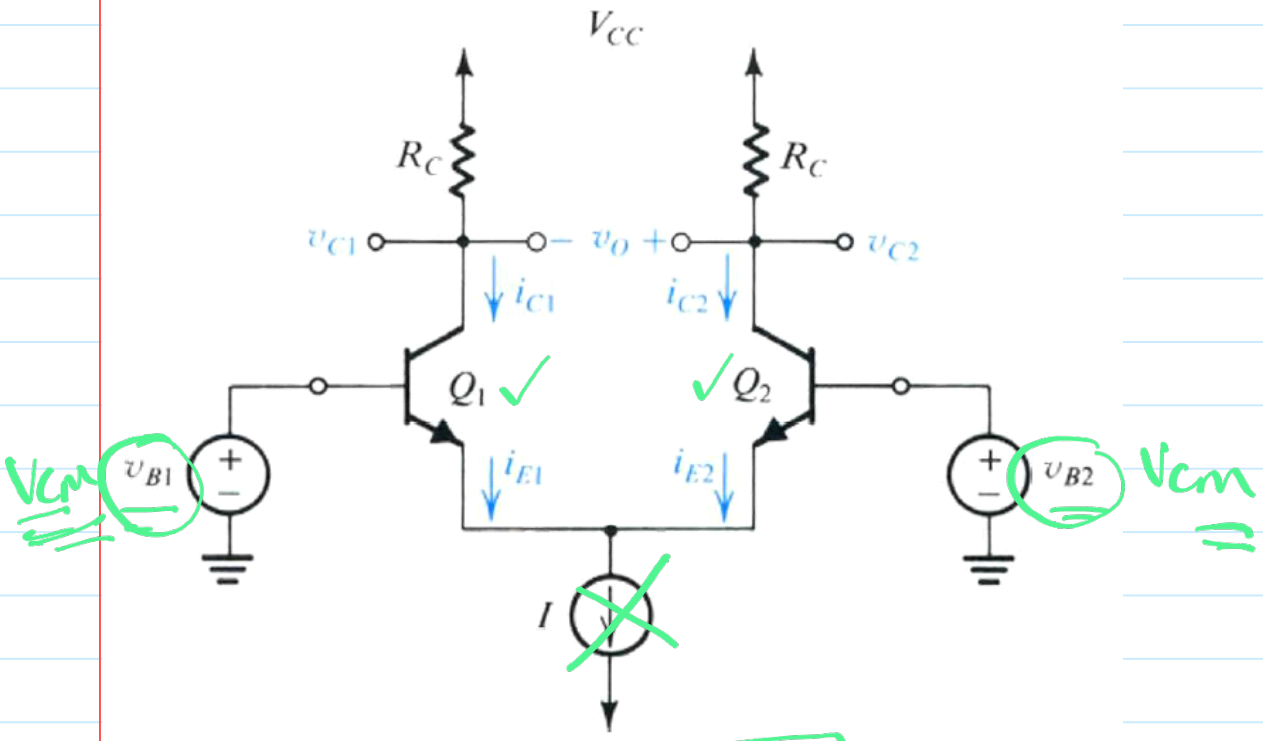


Figure 9.12 (**a**) Differential amplifier with current-source loads formed by Q_3 and Q_4. (**b**) Differential half-circuit of the amplifier in (**a**).

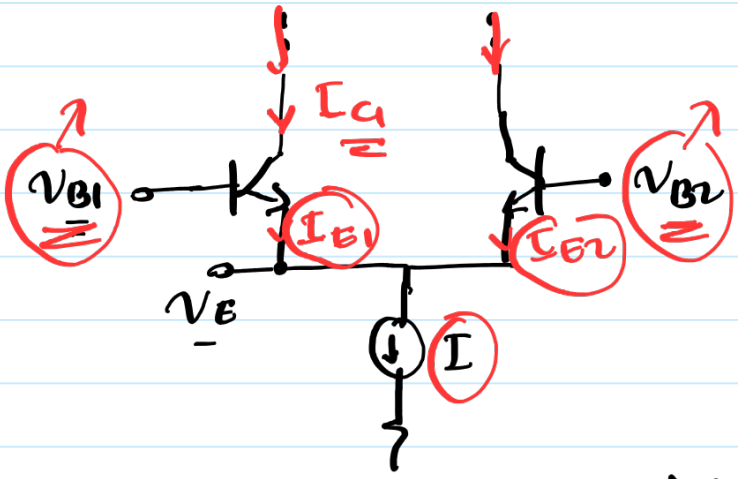
Recap



$$V_{CMmax} \cong V_C + 0.4 = V_{CC} - \alpha \frac{I}{2} R_C + 0.4$$

$$V_{CMmin} = -V_{EE} + V_{CS} + V_{BE}$$

Large Signal Operation



$V_{B2} - V_{B1} = V_{id}$

$$I_{E1} = \frac{I_S}{\alpha} e^{(V_{B1} - V_E)/V_T} \quad \leftarrow \frac{kT}{q}$$

$$I_{E2} = \frac{I_S}{\alpha} e^{(V_{B2} - V_E)/V_T}$$

$$\frac{I_{E1}}{I_{E2}} = e^{(V_{B1} - V_E - V_{B2} + V_E)/V_T}$$

$$\frac{I_{E1}}{I_{E2}} = e^{(V_{B1} - V_{B2})/V_T}$$

$$1 + \frac{I_{E1}}{I_{E2}} = e^{(V_{B1} - V_{B2})/V_T} + 1$$

$$\frac{I_{E2} + I_{E1}}{I_{E2}} = e^{(V_{B1} - V_{B2})/V_T} + 1$$

$$\frac{I_{E2}}{I_{E2} + I_{E1}} = \frac{1}{1 + e^{(V_{B1} - V_{B2})/V_T}}$$

$I_{E2} + I_{E1} = I$

$$I_{E2} = \frac{I}{1 + e^{(V_{B1} - V_{B2})/V_T}}$$

$V_{B1} - V_{B2} = -V_{id}$

$$= \frac{I}{1 + e^{-V_{id}/V_T}}$$

$$I_{E1} = \frac{I}{1 + e^{(V_{B2} - V_{B1})/V_T}}$$

$V_{B2} - V_{B1} = V_{id}$

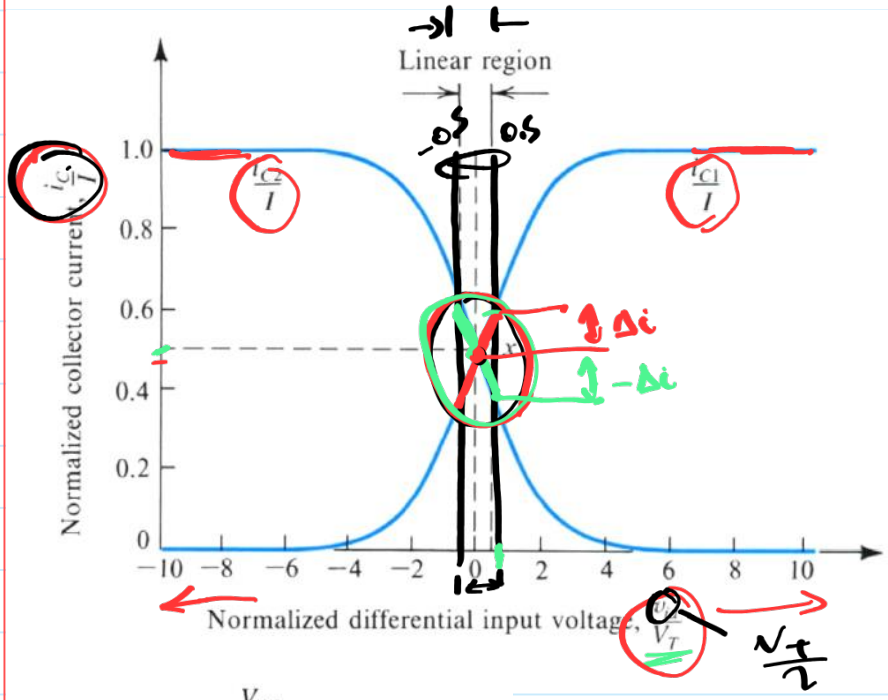
$$= \frac{I}{1 + e^{V_{id}/V_T}}$$

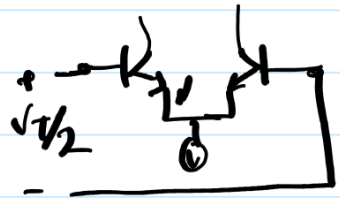
Large Signal Operation

$$\frac{i_{C1}}{I} = \frac{\alpha}{1 + e^{-v_{id}/V_T}}$$

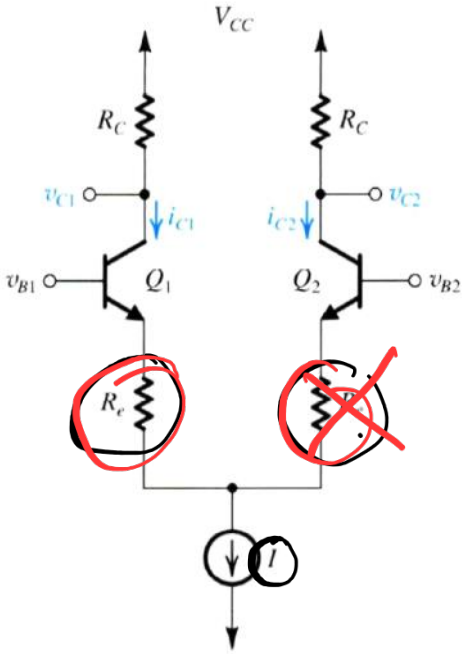
$\alpha \approx 1$

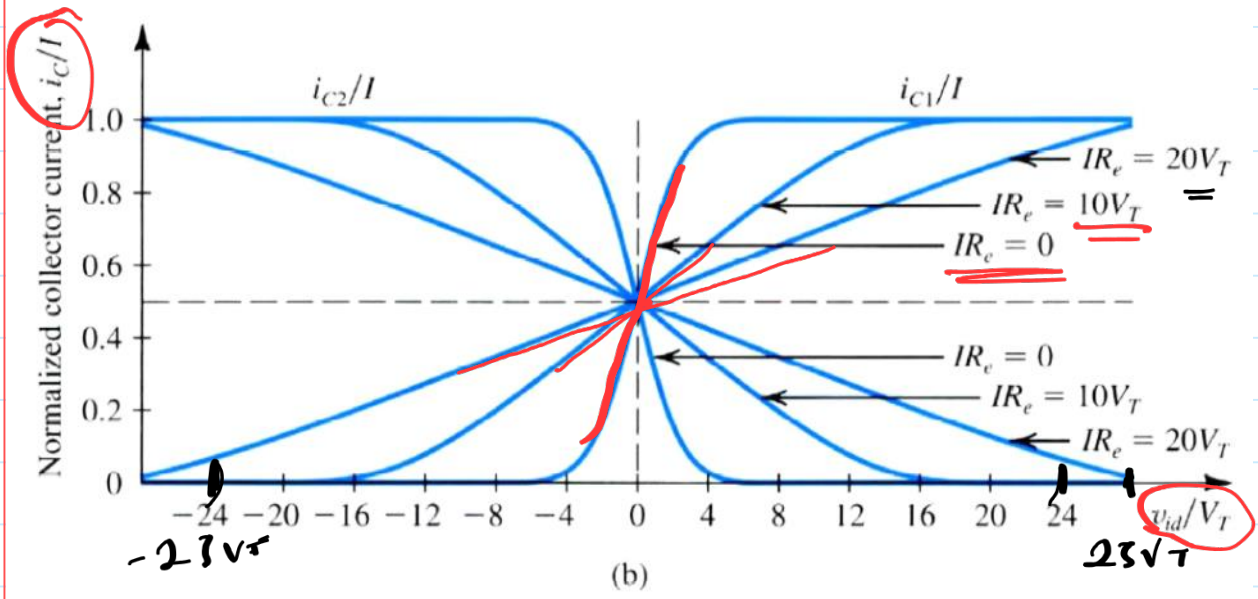
$\approx \frac{1}{2}$





$\frac{25mV}{2} = 12.5mV$





$$i_{C1} = \frac{\alpha I}{1 + e^{-v_{id}/V_T}} \cdot \frac{e^{v_{id}/2V_T}}{e^{v_{id}/2V_T}}$$

$$i_{C2} = \frac{\alpha I}{1 + e^{v_{id}/V_T}}$$

$v_{id} \ll 2V_T$

$\frac{v_{id}}{2V_T} \approx 0$

α

$$i_{C1} = \frac{\alpha . I \; e^{v_{id}/2V_T}}{e^{v_{id}/2V_T} + e^{-v_{id}/2V_T}}$$

$$i_{C1} = \frac{\alpha I \left(1 + v_{id}/2V_T\right)}{1 + v_{id}/2V_T + 1 - v_{id}/2V_T}$$

$$= \frac{(\alpha I)(1 + v_{id}/2V_T)}{2}$$

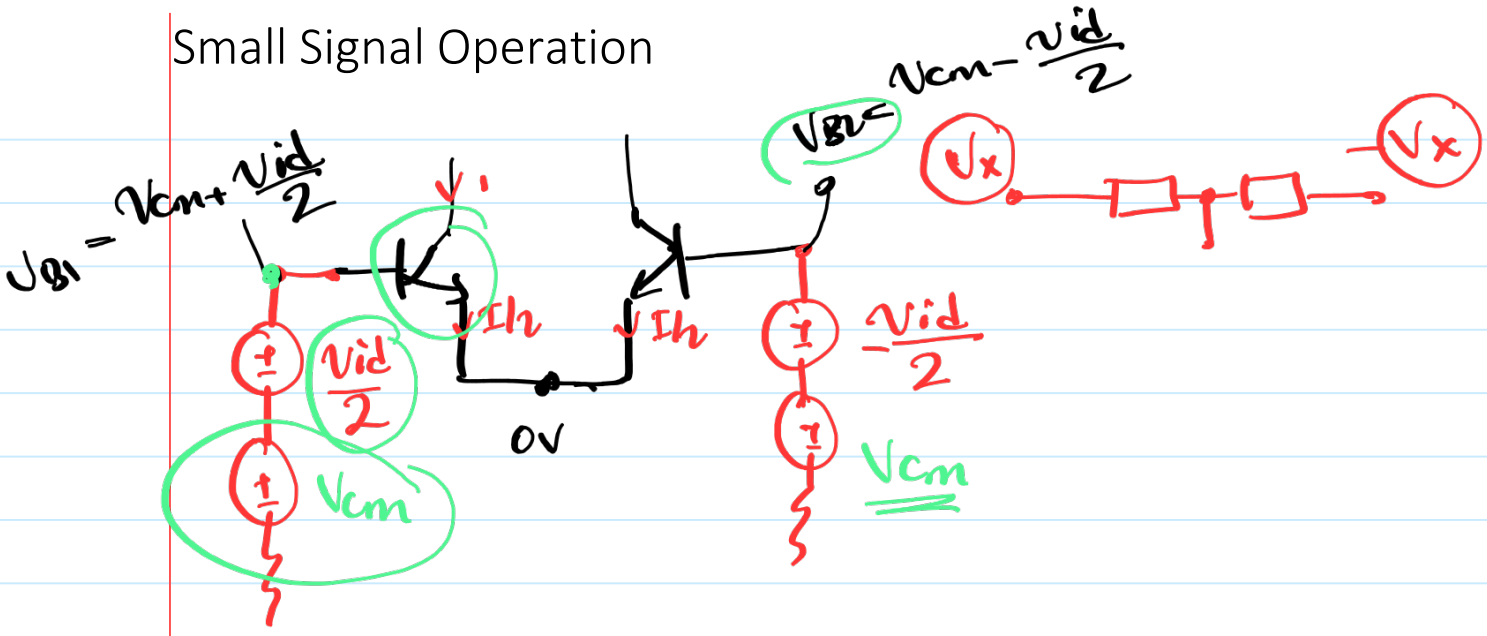
$$i_{C1} = \frac{\alpha I}{2} + \frac{\alpha I}{2V_T} \cdot \frac{v_{id}}{2}$$

$$i_{C2} = \frac{\alpha I}{2} - \frac{\alpha I}{2V_T} \cdot \frac{v_{id}}{2}$$

$$i_C = \frac{\alpha I}{2V_T} \cdot \frac{v_{id}}{2}$$

g_m

Small Signal Operation

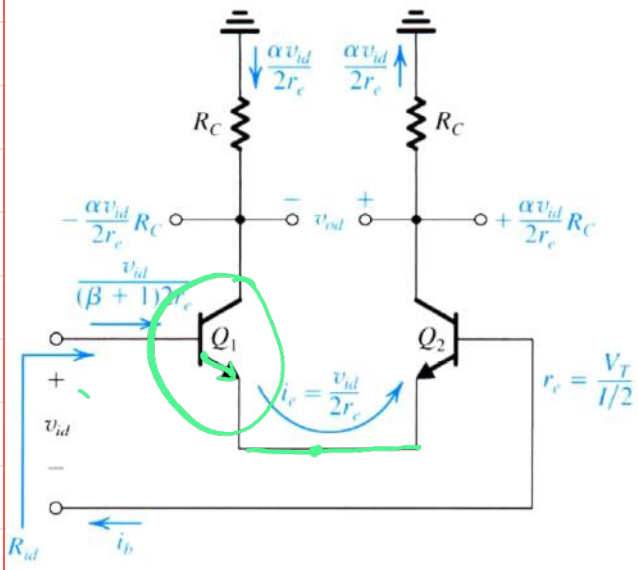


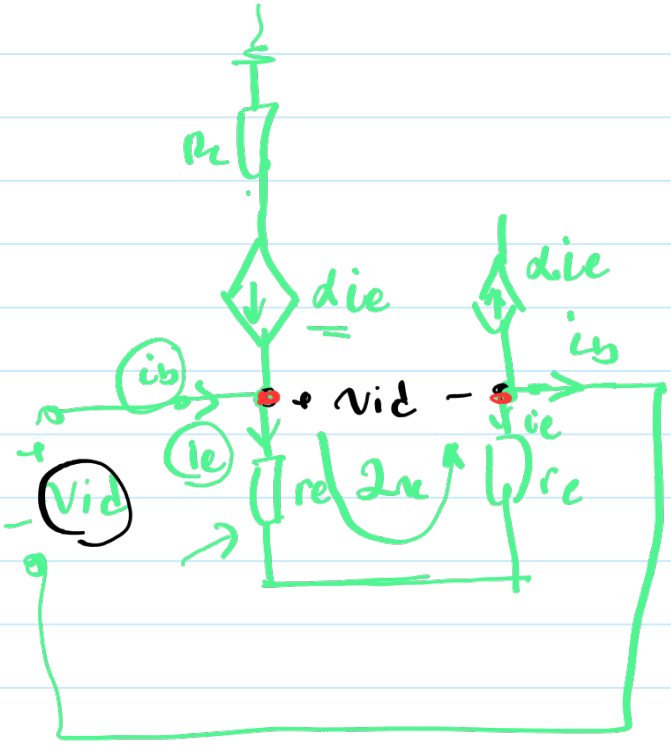
$$v_{BE1} = V_{BE} + \frac{v_{id}}{2} \qquad v_{BE2} = V_{BE} - \frac{v_{id}}{2}$$

$$g_m = \frac{I_C}{V_T} = \frac{\alpha I}{2} / V_T$$

$$\underline{i_c} = g_m \frac{v_{id}}{2}$$

Small Signal Operation

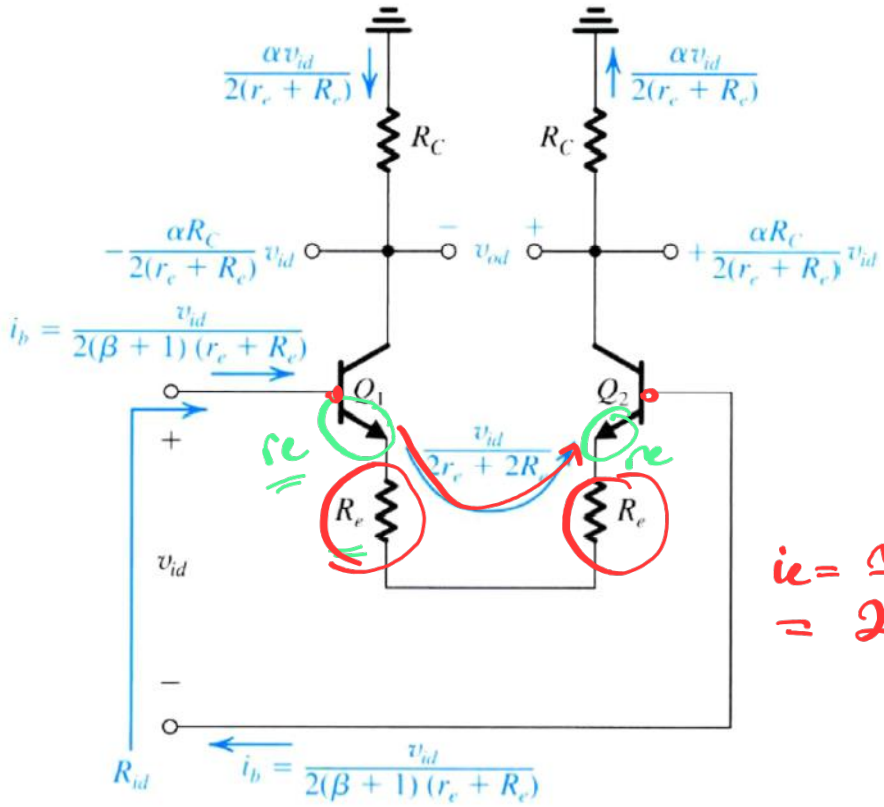




$ie = \frac{vid}{2re} = \boxed{gm \cdot \frac{vid}{2\alpha}}$

α/gm

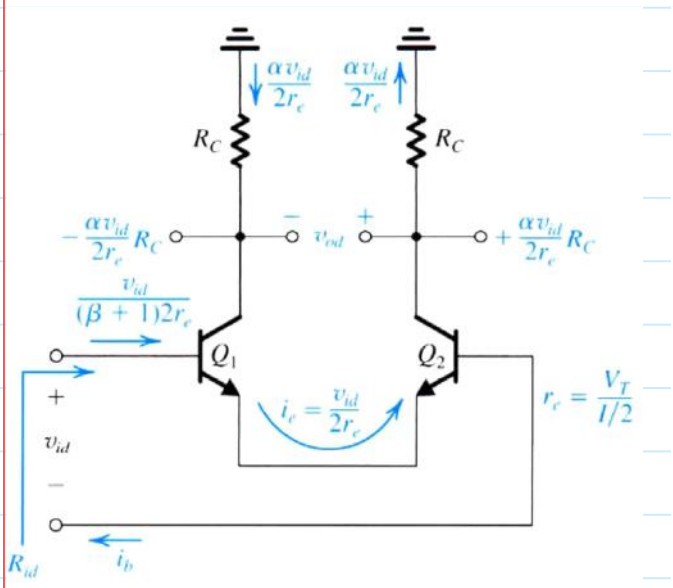
$\alpha.\ ie = \frac{ic}{2} = gm\frac{vid}{2\alpha} \cdot \alpha = gm\frac{vid}{2}$

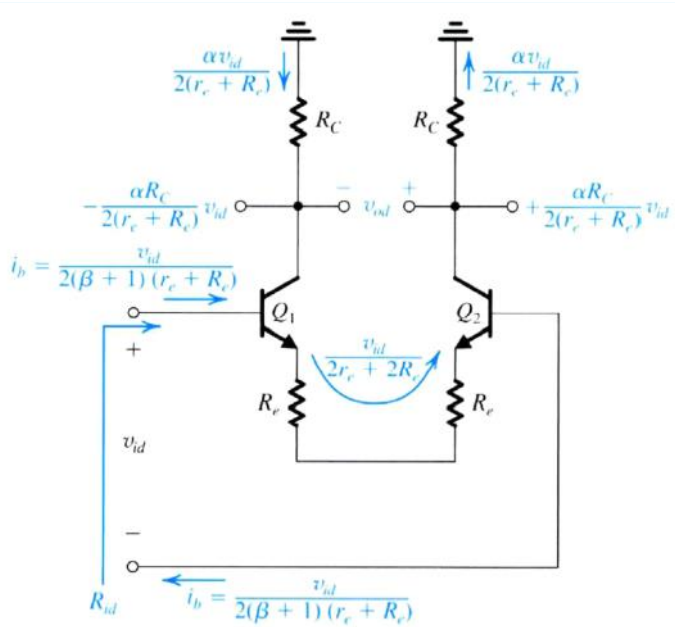


$R = (re + Re)2$

$ie = \frac{vid}{2re + 2Re}$

Input Differential Resistance





Differential Voltage Gain

Differential Half Circuit

Template